

Assignment 4

1. Multiple Choice Questions

1. For a given gas solute in a solvent, Henry's Law constant increases when:
 - ☒ a) Temperature decreases
 - ☐ b) Temperature increases
 - ☐ c) External gas pressure decreases
 - ☐ d) Molar mass increases
2. Which concentration unit is temperature-independent?
 - ☐ a) Molarity
 - ☒ b) Molality
 - ☐ c) Volume percent
 - ☐ d) Mass/volume %
3. A 0.10 m solution of Na_2SO_4 ideally has how many particles?
 - ☐ a) 0.10 m
 - ☐ b) 0.20 m
 - ☒ c) 0.30 m
 - ☐ d) 0.40 m
4. Addition of nonvolatile solute to solvent:
 - ☐ a) Increases vapor pressure
 - ☒ b) Decreases vapor pressure
 - ☐ c) Raises freezing point
 - ☐ d) Lowers boiling point

5. Osmotic pressure (π) formula is:

- ☒ a) $\pi = iMRT$
- ☐ b) $\pi = MRT/i$
- ☐ c) $\pi = iRT/M$
- ☐ d) $\pi = MiRT$

6. Tyndall effect is due to:

- ☐ a) Refraction at solvent surface
- ☐ b) Diffraction of X-rays by solute
- ☒ c) Scattering of light by colloidal particles
- ☐ d) Formation of micelles

7. Which lowers gas solubility in liquid most?

- ☐ a) Decreasing T
- ☐ b) Increasing P
- ☒ c) Increasing T
- ☐ d) Adding salt

8. The K_f of benzene is $5.12\text{ }^\circ\text{C}\cdot\text{kg/mol}$. Adding 0.500 mol nonvolatile solute to 2.00 kg benzene lowers freezing point by:

- ☐ a) $2.56\text{ }^\circ\text{C}$
- ☒ b) $1.28\text{ }^\circ\text{C}$
- ☐ c) $5.12\text{ }^\circ\text{C}$
- ☐ d) $0.128\text{ }^\circ\text{C}$

9. At constant T, osmotic pressure is proportional to:

- ☒ a) Molarity
- ☐ b) Molality
- ☐ c) Mole fraction
- ☐ d) $1/\text{Molarity}$

10. For the reaction $A \rightarrow \text{products}$, doubling [A] causes the rate to quadruple. The order with respect to A is:

- ☐ a) 0

- ☐ b) 1
- ☒ c) 2
- ☐ d) 4

11. The rate law for a second-order reaction (single reactant) is:

- ☐ a) $\text{Rate} = k[A]$
- ☒ b) $\text{Rate} = k[A]^2$
- ☐ c) $\text{Rate} = k/[A]^2$
- ☐ d) $\text{Rate} = k \ln[A]$

12. Units of k for a first-order reaction are:

- ☐ a) $\text{M}^{-1} \cdot \text{s}^{-1}$
- ☐ b) $\text{M}^{-2} \cdot \text{s}^{-1}$
- ☒ c) s^{-1}
- ☐ d) $\text{M} \cdot \text{s}^{-1}$

13. For a first-order process, half-life depends on:

- ☐ a) Initial concentration only
- ☒ b) k only
- ☐ c) Both a and b
- ☐ d) Neither

14. The integrated rate law for a zero-order reaction is:

- ☒ a) $[A] = -kt + [A]_0$
- ☐ b) $1/[A] = kt + 1/[A]_0$
- ☐ c) $\ln[A] = -kt + \ln[A]_0$
- ☐ d) $\ln t = [A]_0 - kt$

15. For a reaction with rate = $k[A][B]$, doubling both $[A]$ and $[B]$ increases rate by:

- ☐ a) $2\times$
- ☒ b) $4\times$
- ☐ c) $8\times$
- ☐ d) No change

16. The slope of $\ln k$ vs $1/T$ plot equals:

- ☒ a) $-E_a/R$
- ☐ b) E_a/R
- ☐ c) k
- ☐ d) $-k$

17. In Arrhenius equation $k = A e^{(-E_a/RT)}$, A is:

- ☐ a) Activation energy
- ☒ b) Pre-exponential factor
- ☐ c) Reaction order
- ☐ d) Gas constant

18. A mechanism step with 3 molecules colliding is called:

- ☐ a) Unimolecular
- ☐ b) Bimolecular
- ☒ c) Termolecular
- ☐ d) Trimolecular

19. A catalyst increases reaction rate by:

- ☐ a) Increasing E_a
- ☒ b) Lowering E_a
- ☐ c) Increasing ΔG°
- ☐ d) Increasing ΔH°

20. In a fast equilibrium followed by slow step, the rate law:

- ☐ a) Is always second order
- ☒ b) Is determined by slow step

- c) Is determined by fast step
- d) Depends only on temperature

21. Plot of $1/[A]$ vs t is linear. What is the order?

- a) 0
- b) 1
- c) 2
- d) Cannot tell

22. Which condition increases rate most for endothermic reaction?

- a) Lower T
- b) Higher T
- c) Removing catalyst
- d) Removing reactant

23. The minimum energy needed for a reaction to occur is:

- a) E_a
- b) A
- c) ΔH
- d) ΔG

24. For an enzyme-catalyzed reaction, raising T too much:

- a) Speeds rate indefinitely
- b) Denatures enzyme, lowering rate
- c) Lowers E_a more
- d) Has no effect

25. For a first-order decay, 75% decomposed means elapsed time is:

- a) $t_{1/2}$
- b) $2 t_{1/2}$

○ c) $t_{1/2} / 2$

○ d) $3 t_{1/2}$

26. A mechanism inconsistent with rate law is:

- a) Valid
- **b) Invalid**
- c) Possible
- d) Elementary

27. For an elementary step $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$, molecularity is:

- a) Uni
- **b) Bi**
- c) Tri
- d) Zero

2. Short-Answer/Calculation Questions

1. A sugar solution has vapor pressure 22.4 torr at 25 °C; pure water has 23.8 torr. Find mole fraction of sugar. $X_{\text{sugar}} = 0.059$

$$X_{\text{water}} = \frac{n_{\text{sol}}}{n_{\text{total}}} = \frac{P_{\text{sol}}}{P_{\text{total}}} = \frac{22.4}{23.8} = 0.941$$
2. Calculate molality of 15.0 g NaCl in 250.0 g H₂O.

$$m = \frac{0.256 \text{ mol NaCl}}{0.250 \text{ kg H}_2\text{O}} = 1.025 \text{ m}$$
3. Solubility of O₂ at 25 °C and 1.0 atm = 1.3×10^{-3} M. Find solubility at 3.0 atm.

$$k = \frac{S_1}{P_1} = \frac{S_2}{P_2} \rightarrow \frac{1.3 \times 10^{-3}}{1} = \frac{S_2}{3} \rightarrow S_2 = 3.9 \times 10^{-3} \text{ M}$$
4. Osmotic pressure of 0.010 M protein solution at 25 °C is 0.225 atm. Find molar mass. $\pi = iMRT = i \frac{D}{\mu} RT$

$$\mu = \frac{iDRT}{\pi} = \frac{1 \cdot (50 \text{ g/L}) \cdot (0.0821 \text{ L}\cdot\text{atm/mol}\cdot\text{K}) \cdot (298 \text{ K})}{0.225 \text{ atm}} = 5436.84 \text{ g/mol}$$
5. Which would have largest freezing point depression: 0.10 m NaCl, 0.10 m CaCl₂, 0.10 m glucose?

$$i=2 \quad i=3 \quad i=1$$

$$0.10 \text{ m CaCl}_2 \quad (\text{m equal, } K_b \text{ equal} \rightarrow \text{depend on } i)$$
6. $A \rightarrow \text{products}$ is first order; $k = 0.115 \text{ s}^{-1}$. Find $t_{1/2}$. $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.115} = 6.026 \text{ s}$
7. $0.500 \text{ M } A \rightarrow \text{products}$ (first order), $k = 0.693 \text{ min}^{-1}$. Time for [A] to reach 0.0625 M?

$$t_{1/2} = \frac{0.693}{0.693 \text{ min}^{-1}} = 1 \text{ min}$$

$$\frac{0.500}{0.0625} = 8 \rightarrow \text{reach half-life 4 times}$$

$$\therefore \text{Time used} = 1 \text{ min} \times 4 = 4 \text{ minutes}$$

8. For second order reaction ($k = 0.250 \text{ M}^{-1} \cdot \text{min}^{-1}$) from $[A]_0 = 0.200 \text{ M}$ to 0.0500 M , $t = ?$

$$t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{(0.250 \text{ M}^{-1} \cdot \text{min}^{-1})(0.200 \text{ M})} = 20 \text{ min}$$

To reach 0.0500 M , it requires 2 periods of half life $\rightarrow$ time taken $= 20 \text{ min} \times 2 = 40 \text{ min}$

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$
$$20 - 5 = 0.250 \cdot t \rightarrow t = 60 \text{ min}$$